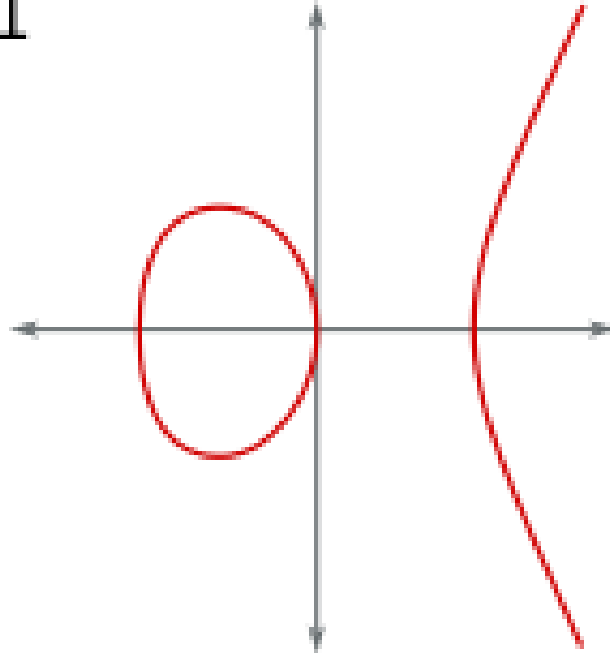


Cryptography and Network Security

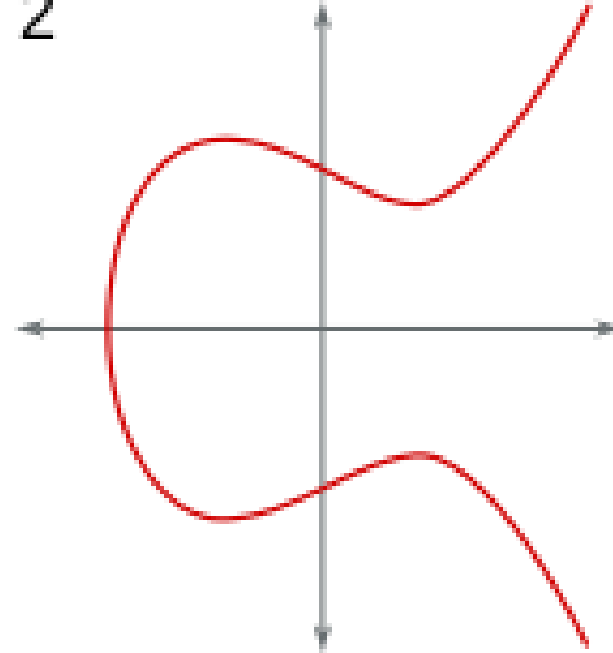
Elliptic Curve Cryptography

1



$$y^2 = x^3 - x$$

2



$$y^2 = x^3 - x + 1$$

What's wrong with RSA?

- **Relies on a belief**
 - Factoring large numbers is *assumed* hard, but **never proven**
- **Prime numbers keep growing**
 - To stay secure, primes must get larger and larger
- **Factoring research is active**
 - Constant progress means today's RSA may be broken tomorrow
- **Quantum computing is a threat**
 - Could **break RSA instantly** using Shor's algorithm

What exactly is an elliptic curve?

- **The Equation:**

$$y^2 = x^3 + ax + b$$

- a and b are constants (real numbers)
- Condition: $4a^3 + 27b^2 \neq 0 \rightarrow$ ensures the curve is **smooth** (no cusps or self-intersections)

The Set of Points:

- All (x, y) points that satisfy the equation
- Plus one extra point:
O = "point at infinity"
(Think of it as a virtual point added for mathematical completeness)

Why It Matters:

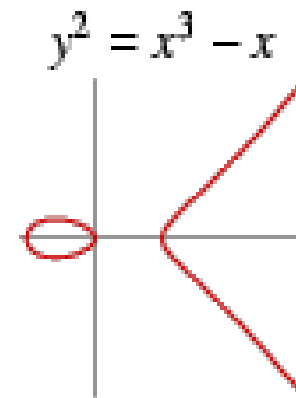
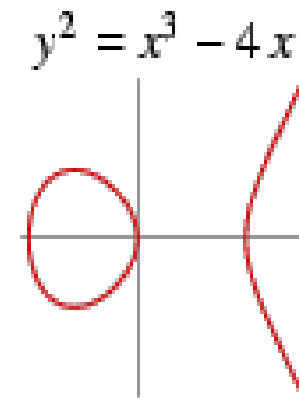
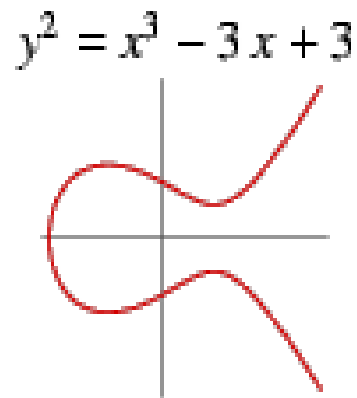
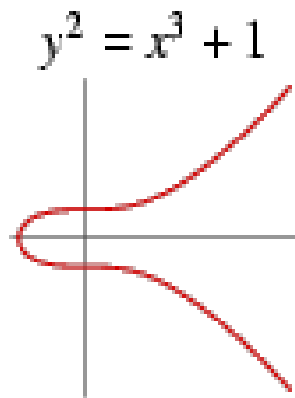
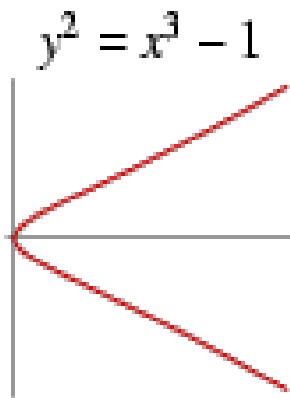
- This curve is **non-singular** $\rightarrow$ smooth and well-behaved
- Used in **cryptography** (e.g., ECC) because points on the curve can be added together to form a group

General form of a EC

- An *elliptic curve* is a plane curve defined by an equation of the form

$$y^2 = x^3 + ax + b$$

Examples



Singular Elliptic Curve

- If $4a^3 + 27b^2 = 0$, then we have a *singular elliptic curve*.
- This could potentially lead to having to not having 3 distinct roots.
- Therefore, we must deal with non-singular elliptic curves with the condition $4a^3 + 27b^2 \neq 0$, in order to assure that we have 3 distinct roots.
- This will allow us to establish the fact that the solution set E forms an Abelian group.

What is a Group?

- Suppose we have any binary operation, such as addition (+), that is defined for every element in a *set* G , which is denoted $(G, +)$
- Then G is a *group* with respect to addition if the following conditions hold:
 - 1.) **G is *closed* under addition:** $x \in G, y \in G$,
imply $x + y \in G$
 - 2.) **$+$ is associative.** For all $x, y, z, \in G$,
$$x + (y + z) = (x + y) + z$$
 - 3.) **G has an identity element e .** There is an e in G such that $\mathbf{x + e = e + x = x}$ for all $\mathbf{x \in G}$.
 - 4.) **G contains inverses.** For each $\mathbf{x \in G}$, there exists $\mathbf{y \in G}$, such that $\mathbf{x + y = y + x = e}$.

What is an Abelian Group

- An Abelian group contains all the rules of a group, but also must meet the following criteria:

5.) **+** is commutative. For all $x \in G$, $y \in G$, $x + y = y + x$.

3 Cases for Solutions

- Suppose $P, Q \in E$, where $P = (x_1, y_1)$ and $Q = (x_2, y_2)$, we must consider three cases:
 - 1.) $x_1 \neq x_2$
 - 2.) $x_1 = x_2$ and $y_1 = -y_2$
 - 3.) $x_1 = x_2$ and $y_1 = y_2$
- These cases must be considered when defining “addition” for our solution set

Defining Addition on E : Case 1

For the case $x_1 \neq x_2$, addition is defined as follows:

$(x_1, y_1) + (x_2, y_2) = (x_3, y_3) \in E$ where

$$x_3 = \lambda^2 - x_1 - x_2$$

$$y_3 = \lambda(x_1 - x_3) - y_1, \text{ and}$$

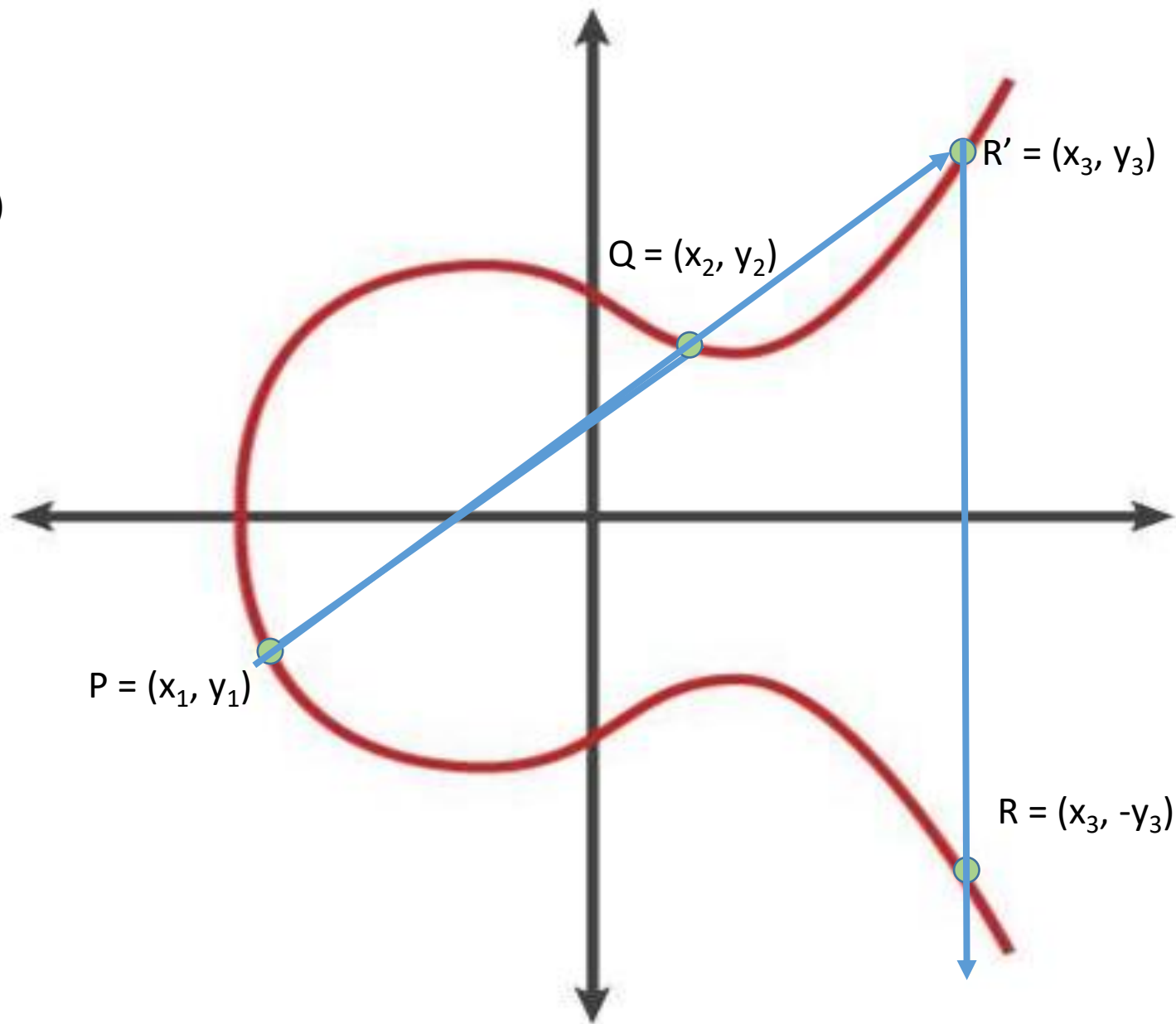
$$\lambda = (y_2 - y_1) / (x_2 - x_1)$$

https://www.umsl.edu/~siegelj/information_theory/projects/elliptic_curves_group_law.pdf

<https://blog.cloudflare.com/a-relatively-easy-to-understand-primer-on-elliptic-curve-cryptography/>

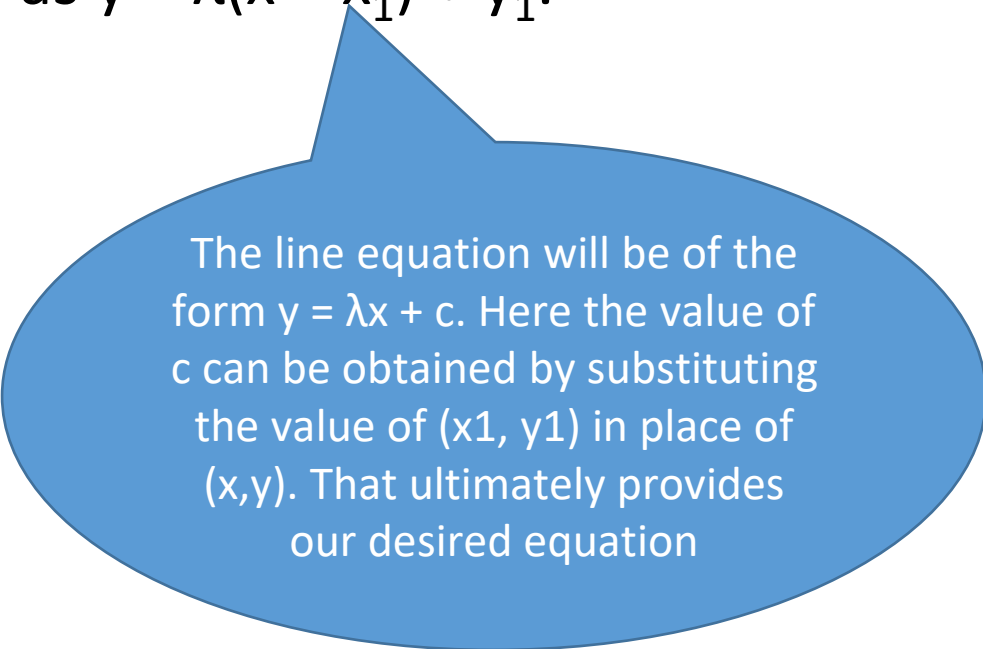
$$P + Q + R' = O$$

$$P + Q = R = (-R')$$



How to get the addition equation

- We know that for a line passing through two points (x_1, y_1) and (x_2, y_2) , we can obtain its slope as $\lambda = (y_2 - y_1) / (x_2 - x_1)$
- Using the slop, we can create the line equation as $y = \lambda(x - x_1) + y_1$.



The line equation will be of the form $y = \lambda x + c$. Here the value of c can be obtained by substituting the value of (x_1, y_1) in place of (x, y) . That ultimately provides our desired equation

How to get the addition equation

- Our task is to identify the intersection point of line and elliptic curve.
- So, we need to identify the point (x_3, y_3) that satisfies the equation
 - $y = \lambda(x - x_1) + y_1$ Eq(1)
 - $y^2 = x^3 + ax + b$ Eq(2)
- Substituting the value of y from eq (1) into eq (2), we will get,

$$y^2 = x^3 + ax + b$$

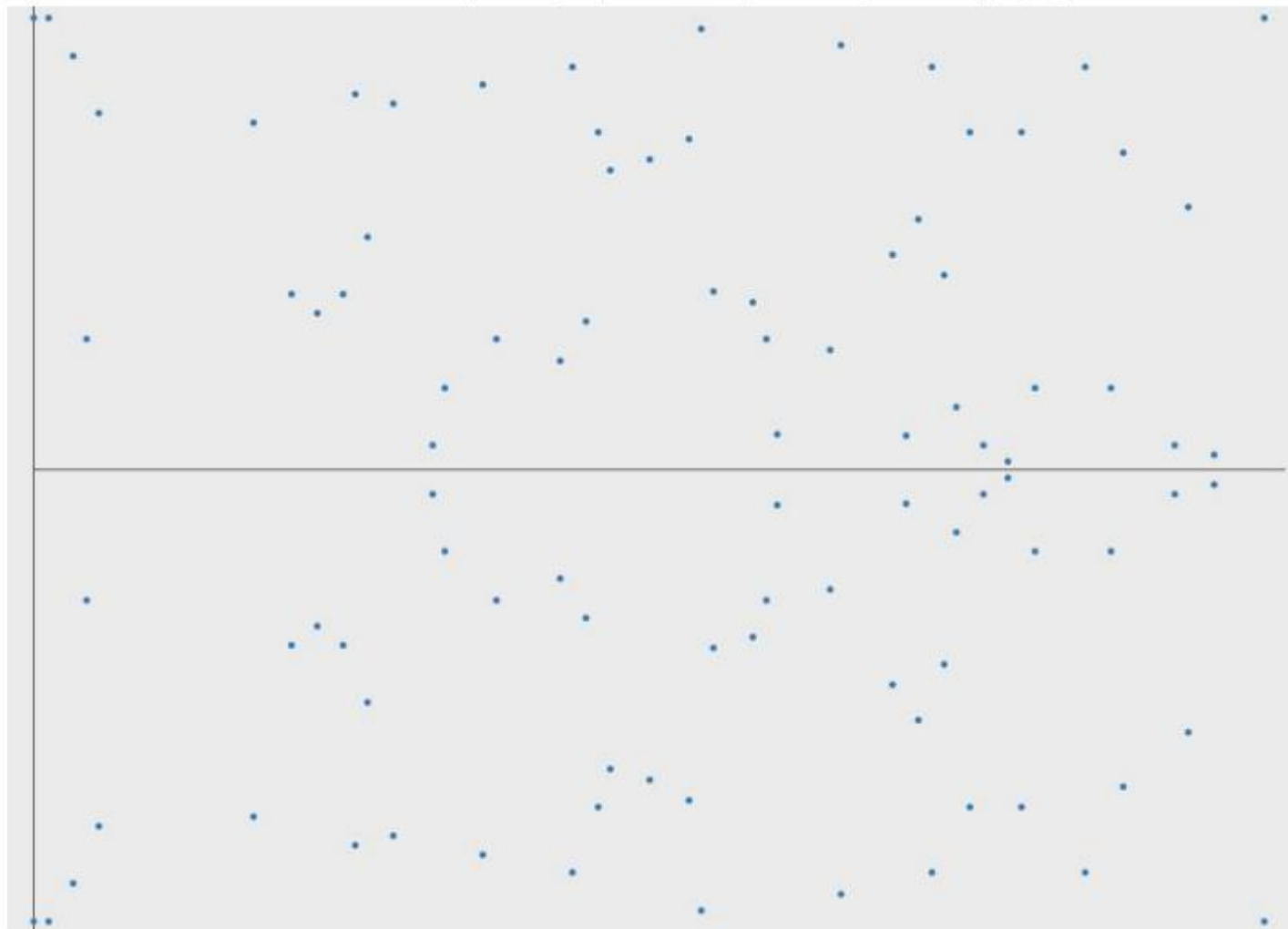
$$(\lambda(x - x_1) + y_1)^2 = x^3 + ax + b$$

$$x^3 - \lambda^2 x^2 + (a + 2\lambda^2 x_1 - 2\lambda y_1)x + (b - \lambda^2 x_1^2 + 2\lambda x_1 y_1 + y_1^2) = 0.$$

How to get the addition equation

- $x^3 - \lambda^2 x^2 + (a + 2\lambda^2 x_1 - 2\lambda y_1)x + (b - \lambda^2 x_1^2 + 2\lambda x_1 y_1 + y_1^2) = 0$Eq (4)
- It is of equation of type $x^3 + Ax^2 + Bx + C = 0$ Eq (5)
- **Let there are three roots r, s, t that solves above equation, then**
 $(x - r)(x - s)(x - t) = 0 \Rightarrow x^3 - (r + s + t)x^2 + \dots$ Eq (6)
- From eq (5) and (6) we can say that $A = -(r + s + t) \Rightarrow t = -A - r - s$
- From eq (4) and (5) we can say that $A = -\lambda^2$
- We already know two solutions x_1 and x_2 , and we need to find third solution x_3 .
Let $r = x_1, s = x_2$, then $t = x_3 = -A - r - s = \lambda^2 - x_2 - x_1$

Figure 1: Scatter plot of $\log_{10}(\text{count})$ vs $\log_{10}(\text{count})$



Defining Addition on E : Case 2

For the case $x_1 = x_2$ and $y_1 = -y_2$, addition is defined as follows:

$$(x_1, y_1) + (x_2, y_2) = (x_3, y_3) \in E \text{ where}$$

$$(x, y) + (x, -y) = O, \text{ the point at infinity}$$

Defining Addition on E : Case 3

For the case $x_1 = x_2$ and $y_1 = y_2$, addition is defined as follows:

$$(x_1, y_1) + (x_2, y_2) = (x_3, y_3) \in E \text{ where}$$

$$x_3 = \lambda^2 - x_1 - x_2 = \lambda^2 - 2x_1$$

$$y_3 = \lambda(x_1 - x_3) - y_1, \text{ and}$$

$$\lambda = (3x_1^2 + a) / 2y_1$$

Defining the Identity

- The point at infinity O , is the identity element. $P + O = O + P = P$, for all $P \in E$.
- From Case 2, and the Identity Element, we now have the existence of inverses.
- Beyond the scope here to prove that we have commutativity and associativity as well.
- Therefore the set of solutions E , forms an Abelian group.

Elliptic Curves modulo p

- Let $p > 3$ be prime. The elliptic curve $y^2 = x^3 + ax + b$ over $\mathbb{Z}_p$ is the set of solutions $(x,y) \in \mathbb{Z}_p \times \mathbb{Z}_p$ to the congruence:

$$y^2 \equiv x^3 + ax + b \pmod{p}$$

where $a \in \mathbb{Z}_p$, $b \in \mathbb{Z}_p$, are constants such that $4a^3 + 27b^2 \not\equiv 0 \pmod{p}$, together with a special point O called the *point at infinity*.

- Solutions still form an Abelian group

So now for an example

- Let's examine the following elliptic curve as an example:

$$y^2 = x^3 + x + 6 \text{ over } \mathbb{Z}_{11}$$

X	0	1	2	3	4	5	6	7	8	9	10
$x^3 + x + 6 \bmod 11$	6	8	5	3	8	4	8	4	9	7	4
QR?	N	N	y	y	N	y	N	y	y	N	y
Y			4,7	5,6		2,9		2,9	3,8		2,9

Generating our group

- From the previous chart, and including the point at infinity O , we have a group with 13 points.
- Since the $O(E)$ is prime, the group is cyclic.
- We can generate the group by choosing any point other than the point at infinity.
- Let our generator = $\alpha = (2,7)$

$$y^2 = x^3 + x + 6 \text{ over } \mathbb{Z}_{11}$$

The Group

For P+P

- $x_3 = \lambda^2 - 2x_1$
- $y_3 = \lambda(x_1 - x_3) - y_1$
- $\lambda = (3x_1^2 + a) / 2y_1$

We can generate this by using the rules of addition we defined earlier where $2\alpha = \alpha + \alpha$

$$\alpha = (2,7); 2\alpha = (5,2); 3\alpha = (8,3); 4\alpha = (10,2);$$

$$5\alpha = (3,6); 6\alpha = (7,9); 7\alpha = (7,2); 8\alpha = (3,5)$$

$$9\alpha = (10,9); 10\alpha = (8,8); 11\alpha = (5,9); 12\alpha = (2,4)$$

For P+Q

- $x_3 = \lambda^2 - x_1 - x_2$
- $y_3 = \lambda(x_1 - x_3) - y_1$
- $\lambda = (y_2 - y_1) / (x_2 - x_1)$

Encryption Rules

- Suppose we let $\alpha = (2,7)$ and choose the private key to be 7
- then $\beta = 7\alpha = (7,2)$
- Encryption:
$$e_k(x,k) = (k(\alpha), x + k(\beta))$$
$$e_k(x,k) = (k(2,7), x + k(7,2)) ,$$

where $x \in E$ is *plaintext* and $0 \leq k \leq 12$ is a random value.

Decryption Rule

- Decryption:

$$d_K(y_1, y_2) = y_2 - K_{\text{priv}} y_1$$

$$d_K(y_1, y_2) = y_2 - 7y_1$$

- This is based on the ElGamal scheme of elliptic curve encryption

Using this Scheme

- Suppose Alice wants to send a message to Bob.
- Plaintext is $x = (10,9)$ which is a point in E
- Choose a random value for k , $k = 3$
- So now calculate (y_1, y_2) :
- $y_1 = 3(2,7) = (8,3)$
- $y_2 = (10,9) + 3(7,2) = (10,9) + (3,5) = (10,2)$
- Alice transmits $y = ((8,3),(10,2))$

Encryption of x
$$e_k(x,k) = (y_1, y_2)$$
$$=(k(\alpha), x + k(\beta))$$

Bob Decrypts

- Bob receives $y = ((8,3),(10,2))$
- Calculates

$$\begin{aligned}x &= (10,2) - 7(8,3) \\&= (10,2) - (3,5) \\&= (10,2) + (3,6) \\&= (10,9)\end{aligned}$$

Which was the plaintext

Decryption:

$$d_k(y_1, y_2) = y_2 - K_{\text{priv}} y_1$$

Problem for student

1. Show that the condition $4a^3 + 27b^2 \not\equiv 0 \pmod{p}$ is fulfilled for the curve $y^2 \equiv x^3 + 2x + 2 \pmod{17}$
2. Let E be an elliptic curve defined over $\mathbb{Z}_7$:
$$E : y^2 = x^3 + 3x + 2.$$
 - I. Compute all points on E over $\mathbb{Z}_7$.
 - II. What is the order of the group?

Elliptic Curve Cryptography

Optimization of Operations

Scalar Multiplication: MSB first (Double and Add)

- Input: elliptic curve E together with an elliptic curve point P , a scalar $d = \sum_{i=0}^t d_i 2^i$ with $d_i \in 0,1$ and $d_t = 1$
- Output: $Q=dP$
- Initialization: $Q=P$
- Algorithm:
 - For $i=t-1$ to 0
 - $Q=2Q \bmod n$
 - If $d_i=1$ then
 - $Q=Q+P \bmod n$
 - End if
 - End for
 - Return Q

Sequential Algorithm

Requires m point doublings and $(m-1)/2$ point additions on the average

Example

- **Compute 7P:**

- $7 = (111)_2$
- $7P = 2(2(P) + P) + P \Rightarrow$ 2 iterations are required
- Principle: First double and then add (accumulate)

- **Compute 6P:**

- $6 = (110)_2$
- $6P = 2(2(P) + P)$

Last Lecture's Example

$$y^2 = x^3 + x + 6 \text{ over } \mathbb{Z}_{11}$$

We have calculated following points :

$$P = (2,7)$$

$$2P = (5,2)$$

$$3P = (8,3)$$

$$4P = (10,2)$$

$$5P = (3,6)$$

$$6P = (7,9)$$

$$7P = (7,2)$$

$$8P = (3,5)$$

$$9P = (10,9)$$

$$10P = (8,8)$$

$$11P = (5,9)$$

$$12P = (2,4)$$

For P+P

- $x_3 = \lambda^2 - 2x_1$
- $y_3 = \lambda(x_1 - x_3) - y_1$
- $\lambda = (3x_1^2 + a) / 2y_1$

For P+Q

- $x_3 = \lambda^2 - x_1 - x_2$
- $y_3 = \lambda(x_1 - x_3) - y_1$
- $\lambda = (y_2 - y_1) / (x_2 - x_1)$

Let's calculate $7P$ in an Optimized way

1. $Q = P = (2,7)$
2. $d_2 = 1, d_1 = 1$ and $d_0 = 1$ (Because $7 = (111)_2$)
3. First Iteration : $Q = 2Q = 2(2,7) = (5,2)$
4. $d_1 = 1$, therefore $Q = Q + P = (5,2) + (2,7) = (8,3)$
5. Second Iteration : $Q = 2Q = 2(8,3) = (7,9)$
6. $d_0 = 1$, therefore $Q = Q + P = (7,9) + (2,7) = (7,2)$

Scalar Multiplication: LSB first (Add and Double)

- Input: elliptic curve E together with an elliptic curve point P , a scalar $d = \sum_{i=0}^t d_i 2^i$ with $d_i \in \{0,1\}$ and $d_t = 1$
- Output: $Q=dP$
- Initialization: $Q = 0, R = P$
- Algorithm:
 - For $l = 0$ to $t-1$
 - If $d_l=1$ then
 - $Q=Q+R \bmod n$
 - End if
 - $R=2R \bmod n$
 - End for
 - Return $(Q + R) \bmod n$

Can **Parallelize**...

What you are doubling and what you are accumulating are different...

On the average $m/2$ point Additions and $m/2$ point doublings

Let's calculate 7P in an Optimized way

1. $Q = O, R = P = (2,7)$
2. $d_2 = 1, d_1 = 1$ and $d_0 = 1$ (Because $7 = (111)_2$)

First Iteration :

3. $d_0 = 1$, therefore $Q = Q + R = O + (2,7) = (2,7)$
4. $R = 2R = 2(2,7) = (5,2)$

Second Iteration :

5. $d_1 = 1$, therefore $Q = Q + R = (2,7) + (5,2) = (8,3)$
6. $R = 2R = 2(5,2) = (10,2)$

At Last Return $(8,3) + (10,2) = (7,2)$

Compute 31P...

31=(11111) ₂	
MSB First	LSB First
1. Q=2P	1. Q=P, R=2P
2. Q=3P	2. Q=3P, R=4P
3. Q=6P	3. Q=7P, R=8P
4. Q=7P	4. Q=15P, R=16P
5. Q=14P	5. Q=31P, R=32P
6. Q=15P	
7. Q=30P	
8. Q=31P	

Solve the Following Problems

- For the same elliptic curve as described in this ppt, and the point $P = (3,5)$ calculate $11P$ using both the approaches (Double and Add; Add and Double). Calculate the total no. of addition operations done in each of this approach.

Some Facts for Elliptic Curve Group

- The order of an elliptic curve group is the number of elements on that elliptic curve. (The number of points on an elliptic curve over Finite Field F_q is denoted as $\#E(F_q)$)
- An order of an element P , denoted as n , is such that $nP = O$.
- A Primitive element of an elliptic curve group is such that the order of element = order of group.
- If the order of group is prime number, then every element of the group is primitive element.
- The group $E(F_q)$ is cyclic if $\gcd(\#E(F_q), q-1) = 1$.

Hasse's theorem

- Given an elliptic curve E modulo q , the number of points on the curve is denoted by $\#E$ and is bounded by:

$$q+1-2\sqrt{q} \leq \#E(F_q) \leq q+1+ \sqrt{q}.$$

Where:

- q = a prime number (or prime power)
 - $\#E(F_q)$ = total number of points on the curve (including the point at infinity)
-
- **What This Means:**
 - Points are **not random**—they fall within a predictable range
 - The curve has roughly $q + 1$ points, give or take $2\sqrt{q}$
 - Example: If $q = 97$, points are between 78 and 118

Elliptic Curved Discrete Logarithm Problem (ECDLP)

- We have an elliptic curve E
 - P = a starting point (primitive element)
 - T = another point on the curve
 - d = a secret integer (what we want to find)

$$\underbrace{P + P + \dots + P}_{d \text{ times}} = d P = T$$

El-Gamal Digital Signature Scheme

Initialization :

- Select an Elliptic Curve E over a Finite Field F_q and select a primitive element G . Let the order of E is n .

Key Generation :

- Alice selects a random integer k_A from the interval $[1, \dots, n-1]$. The k_A will be her private key.
- She computes $A = k_A G$ and publishes A as her public key.

El-Gamal Digital Signature Scheme

Signature Generation :

1. Alice selects a random integer k from the interval $[1, \dots, n-1]$.
2. Compute $R = kG = (X_R, Y_R)$ and then $r = X_R \bmod n$. If $r = 0$, then goto step 1.
3. Compute $e = H(M)$, where H is a hash function $\{0,1\}^* \rightarrow F_n$
4. Compute $s = k^{-1}(e + rk_A) \bmod n$.
5. Signature for Message M is (R, s)

El-Gamal Digital Signature Scheme

Signature Verification :

- Bob computes $V_1 = sR$.
- Compute $V_2 = h(m)G + rA$. ($r = X_R \bmod n$)
- Accept if and only if $V_1 = V_2$.

El-Gamal Digital Signature Example

Initialization :

- E: $y^2 = x^3 - x + 6 \pmod{29}$
- $G=(5,7)$

Key generation :

- Alice selects a random integer $k_A = 23$
- She computes $A = k_A G = 23(5,7) \pmod{31} = (21,18)$

El-Gamal Digital Signature Scheme

Signature Generation :

1. Selects a random integer $k = 13$.
2. Compute $R = kG = 13(5,7) = (X_R, Y_R) = (7,27)$, Compute $r = X_R \bmod n = 7 \bmod 31 = 7$
3. Let the $e = H(M) = 2$
4. Compute $s = k^{-1}(e + rk_A) \bmod n = 13^{-1} (2 + 7 \cdot 23) \bmod 31 = 3$.
5. Signature for Message M is $(R,s) = ((7,27),3)$.

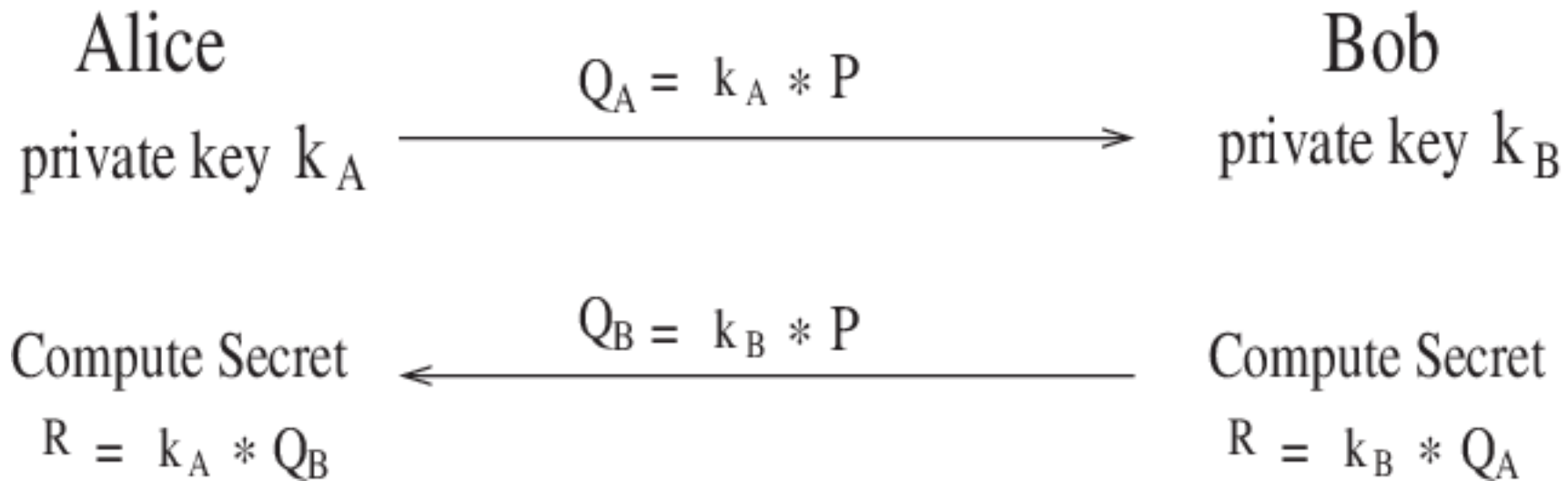
El-Gamal Digital Signature Scheme

Signature Verification :

- Bob computes $V_1 = sR = 3(7,27) \bmod 31 = (21,11)$.
 $(3R = 3 * 13 * G \bmod 31 = 8 * G = (21,11))$
- Compute $V_2 = h(m)G + rA = 2*(5,7) + 7*(21,18) = (21,11)$.
 $(2 * G + 7 * 23 * G \bmod 31 = 2 * G + 6 * G = 8 * G)$
- Accepted because $V_1 = V_2$.

Diffie-Hellman Key Exchange with ECC

DiffieHellman (ECDH)



Problems to solve

- For the elliptic curve $y^2 = x^3 + x + 6$ over $\mathbb{Z}_{11}$ and its primitive root $P = (2,7)$
 1. Create your own private key for El-Gamal digital signature scheme, and for a message whose hash value is 15, generate its signature.
 2. Let this elliptic curve and point P are selected for Diffie-Hellman Key exchange setup. Bob has sent his computed key $Q_B = (7,9)$, and Alice has chosen the private key element as $A = 8$. What will be the shared secret key between Alice and Bob.